Physics 12 Winter 2026
Week 3 Discussion

TA: Jiashu Han

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1. Summary of Topics from Last Week

- *From the lectures*
  - ◇ Momentum $\vec{p} = m\vec{v}$
    - ∗ Faster things have more momentum. More massive (larger m) objects are harder to move (require more momentum).
  - ◇ Conservation of momentum $m_1\vec{v}_1 = m_2\vec{v}_2$
    - ∗ Momentum of a system remains unchanged (initial = final) if no external forces act on the system.
  - ◇ Elastic collision: kinetic energy is conserved
    - ∗ Colliding objects bump off of each other.
  - ◇ Inelastic collision: kinetic energy is not conserved
    - ∗ Colliding objects stick together.
  - ◇ Rotational motion (angular vel. $|\omega| = vr$, angular acc. $|\alpha| = ar$)
  - ◇ Rotational inertia (moment of inertia): $I \propto mr^2$ (depends on geometry)
    - ∗ Rotational "laziness". More massive things are harder to rotate. More spread-out objects are harder to turn.
  - ◇ Torque $|\tau| = r|F|$ (direction based on right-hand rule)
    - ∗ Rotational "force". You can either use a longer lever arm or a larger force to turn an object more easily.
  - ◇ Uniform circular motion, Centripetal force $|F| = \frac{mv^2}{r}$
  - ◇ Angular momentum $\vec{L} = I\vec{\omega}$
    - ∗ Faster rotating things have more angular momentum. More rotationally "lazy" (larger I) objects are harder to rotate (require more angular momentum).
  - ◇ Conservation of angular momentum $I_1\vec{\omega}_1 = I_2\vec{\omega}_2$
    - ∗ Angular momentum of a system remains unchanged (initial = final) if no external torques act on the system.
  - ◇ Kinetic energy KE $= \frac{1}{2}mv^2$, Potential energy
    - ∗ Faster things have more kinetic energy. More massive (larger m) moving objects have more kinetic energy.

- ◇ Work $W = F\Delta x \cos\theta$, Work done **on** object $W = \Delta\text{KE}$
  - ∗ For conservative forces, amount of work depends only on the initial and final states, not the path taken. You can make your life easier by taking advantage of the tradeoff between force and distance.

- *From the textbook* (Wolfson 4th edition)
  - ◇ Forms of energy (mechanical, thermal, electrical, chemical, nuclear, etc.) (first half of Chapter 3)
    - ∗ Make sure you understand the origin and nature of each type of energy we introduced so far (how are they stored?).
  - ◇ Work, Friction (second half of Chapter 3)
  - ◇ Thermal Energy, Heat, Temperature (Chapter 4.1 - 4.3)
  - ◇ Heat Capacity, State Changes, Energy Quality (Chapter 4.4 - 4.6)
  - ◇ Entropy, Laws of Thermodynamics, Heat Engine/Pump (Chapter 4.7 - 4.10)

2. **Momentum & Energy in Action**

(a) ***Collision.** A 1,200 kg car traveling at 20 m/s (approx. 45 mph) rear-ends a stationary 800 kg car. The two cars lock bumpers and move together as one mass.



- Calculate the final velocity v_f of the combined wreck.

  Solution: Since the cars combine into one mass after the collision, the collision is inelastic, so kinetic energy is not conserved. However, we can still use momentum conservation to solve this problem. Let the mass of the heavier car be $M = 1200$ kg and the mass of the lighter car be $m = 800$ kg. The speeds of the cars are $v_1 = 20$ m/s and $v_2 = 0$ m/s, respectively. Momentum conservation then says:

$$Mv_1 + mv_2 = (M + m)v_f \tag{1}$$

  Therefore, $v_f = \frac{Mv_1}{M+m} = 12$ m/s.

- Calculate the total kinetic energy before and after the crash.

  Solution: The total kinetic energy is given by the sum of the kinetic energies of the two cars. Before the collision, the total KE is

$$\text{KE} = \frac{1}{2}Mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}Mv_1^2 = 2.4 \times 10^5 \text{ J} \tag{2}$$

  After the collision, the total KE is

$$\text{KE} = \frac{1}{2}(M+m)v_f^2 = 1.44 \times 10^5 \text{ J} \tag{3}$$

  As expected, the total kinetic energy was not conserved.

- Since energy is "lost" from the motion, where did it go? Name some forms of energy it converted into.

  Solution:

  - ◇ Heat (thermal energy) - Friction and stress cause the microscopic particles to vibrate more rapidly and increase the internal energy. This is still a form of kinetic energy, but for random motion.
  - ◇ Sound energy - Macroscopic vibrations generate sound waves that carry energy away from the system. This is a mix of kinetic and potential energy (the air acts like a spring, which has maximum potential energy and no kinetic energy when fully stretched).
  - ◇ Deformation/Elastic potential energy - Energy is used to either change the shape of the objects permanently (deformation) or temporarily (elastic energy).

(b) **Friction**. A bicycle that travels twice as fast as another when braking to a stop will skid

A. twice as far
B. four times as far
C. eight times as far
D. sixteen times as far
E. depends on the bike's mass

Solution:

The answer depends on what assumptions we make:

- If we make the simplifying (and somewhat unrealistic) assumption that the bike wheels completely lock during braking, and the bike simply skids, then the answer is B.
  For the bike to stop, the work done by friction must be equal to the kinetic energy before braking: $W_f = F_f \Delta x = \frac{1}{2}mv^2$. The force of friction is $F_f = \mu_k F_N = \mu_k mg$ where μ_k is the coefficient of kinetic friction between the tire and the ground. Putting the two equations together, we get $\Delta x = \frac{v^2}{2\mu_k g}$. Assuming the two bikes have the same kind of tire (i.e. same μ_k), this bike will skid four times as far.
- More realistically, the bike wheels don't completely lock, and the bikes don't skid. The friction that dissipates the kinetic energy is between the brake and the wheel, and is independent of the bike's mass. It depends more on how hard you apply the brake. Therefore, the masses don't cancel out in the equation for Δx, and the answer would be E.

3. **Rotational Motion & Torque**

(a) **The Wind Turbine Blade**. A wind turbine's ability to generate power depends on the torque applied by the wind to its blades.

- If the wind exerts a force of 500 N at a distance of 10 m from the hub, what is the torque?

  Solution: The magnitude of the torque is $\tau = rF = 5000\ \text{N} \cdot \text{m}$. Note that torque has the same units as energy (J = N·m), but they are different quantities – torque is a vector and energy is a scalar. We only write the unit for energy as joule.
  The direction of the torque can be determined using the right-hand rule: curl the fingers on your right hand around the direction the blades rotate in, your thumb will then point perpendicular to the plane of the blades. This is the direction of the torque.

- If you replace the blades with much heavier ones of the same length, will it be easier or harder to start the turbine spinning? Why?

  Solution: Recall that the rotational inertia is $I \propto mr^2$, where the proportionality constant depends on the geometry of the object. Since we increase m while keeping r the same, there is more rotational inertia, and it will be harder to turn the blades.

(b) ***Spinning Storm**. If a spinning storm system (like a hurricane or tornado) shrinks in size and becomes narrower, what happens to its rotational speed?

Solution: Unless there is something applying a torque to the storm system, its angular momentum $\vec{L} = I\vec{\omega}$ is conserved. By making the radius of the storm smaller (assuming its mass remains unchanged), we are decreasing the rotational inertia $I \propto mr^2$ of the storm system. Therefore, the rotational speed ω must increase to keep L constant.

4. **Work, Power, and Heat Capacity**

(a) ***Crate on Elevator**. An elevator lifts a 500 kg crate up 20 meters in 10 seconds. Take $g = 10$ m/s^2.

- How much work was done against gravity? What is the change in gravitational potential energy?

  Solution: This is the work done by the elevator, which supplies an upward force $\vec{F}_n$ on the crate. How do we know how much this force is? If we look at the crate itself, it feels two forces: an upward normal/supporting force by the elevator floor (which is $\vec{F}_n$), and a downward gravity $\vec{F}_g = m\vec{g}$. The load remains stationary inside the elevator and does not accelerate relative to the elevator, so the two forces felt by the crate must be equal in magnitude, i.e. $F_n = mg$. The work done by $\vec{F}_n$ is therefore $W = F_n \Delta y \cos 0 = 10^5$ J.
  The change in gravitational potential energy is $\Delta\text{PE} = mg\Delta h = 10^5$ J. This is the same as the work done by the elevator on the crate, regardless of what path the elevator takes, because the force of gravity is a conservative force.

- What was the average power output of the elevator motor in watts?

  Solution: Using the definition of power: $P = \frac{W}{\Delta t} = 10000$ W

- We replace the vertical elevator with a conveyor belt that's tilted at an angle of 30°. What changes now?

  Solution: The force that's responsible for moving the crate is the static friction between the belt and the crate. It's magnitude is $F_f = mg \sin\theta = 2500$ N where $\theta = 30°$ is the angle of the conveyor belt. This force does

work along the conveyor belt, which has a length of $d = \frac{20\text{ m}}{\sin(30°)} = 40$ m. The work done by this force is $W = F_f d = 10^5$ J, same as before! What changed? Because the crate now travels a longer distance, it takes less force to move it up.

- If the motor actually consumed more energy than your answer for work, what is a likely culprit?

  Solution: Non-conservative forces such as friction and air resistance.

(b) ***Electric or Gas Stove?** Currently, most of the electricity we use is generated by fossil fuels, of which the majority comes from natural gas, so let's estimate the energy we consume when boiling 1 gallon (0.0038 m^3) of water. Take the specific heat of water to be 4184 J/(kg·K).

- Assume 100% of the heat generated by the stove gets transferred to the water, how much energy (in the form of heat) does the stove need to generate to bring the water from room temperature (20°C) to boiling (100°)?

  Solution: We use the definition of specific heat:

$$Q = mc\Delta T = 0.0038\text{ m}^3 \times \frac{10^3\text{ kg}}{1\text{ m}^3} \times 4184\text{ J/(kg}\cdot\text{K)} \times 80\text{ K} = 1.3 \times 10^6\text{ J} \tag{4}$$

- Take the heat content of natural gas to be 55 MJ/kg. How much natural gas do we have to burn in a gas stove to generate this much energy?

  Solution:

$$\frac{1.3 \times 10^6\text{ J}}{55\text{ MJ/kg}} = 0.023\text{ kg} \tag{5}$$

- In a natural gas power plant, roughly a third of the thermal energy from burning natural gas gets converted to electricity. Furthermore, electricity can be very efficiently turned into heat by an electric stove. How much natural gas do we have to burn at the power plant to generate the electricity we need to heat up the same water using an electric stove?

  Solution: Let's assume 100% electricity to heat conversion.

$$\frac{1.3 \times 10^6\text{ J}}{33\% \times 55\text{ MJ/kg}} = 0.07\text{ kg} \tag{6}$$

It takes more natural gas to do the same thing! This is because the form of energy we need (heat) is a low quality form of energy, whereas electricity is a high quality form of energy. If low quality energy is all we need, we don't have to go through all the trouble to make it high quality and then turn that back into low quality.

5. **Heat and Thermal Transfer**

   (a) **The Three Modes.** Match the following scenarios to conduction, convection, or radiation:

   - A black asphalt road getting hot on a sunny day: radiation
   - Heat escaping through the glass of a single-pane window: conduction
   - Warm air rising from a heater to the second floor of a house: convection

   (b) **Heat Loss at Home.** The rate of heat conduction through a wall is given by $P = k\frac{A\Delta T}{d}$, where k is the thermal conductivity of the material, ΔT is the difference in temperature across the wall, d and A are the thickness and area of the wall, respectively.

   - Let's assume the wall is 0.3 m (about a foot) thick with an area of 10 m^2 and made of concrete ($k = 1$ W/(m·K)). It's winter in San Diego, so the temperature outside is about 10°C while the temperature inside is a comfortable 20°C. Calculate the heat loss through the wall.

     Solution: We use the equation provided:

     $$P = k\frac{A\Delta T}{d} = 1 \text{ W/(m} \cdot \text{K)}\frac{10 \text{ m}^2 \times 10 \text{ K}}{0.3 \text{ m}} = 333 \text{ W} \tag{7}$$

   - Why is a "dead air space" between two panes of glass a better insulator than a solid sheet of glass?

     Solution: Air has a much lower thermal conductivity ($k = 0.026$ W/(m · K)) than glass ($k = 0.8$ W/(m · K)).